

NEEL PATEL

Python Developer | Machine Learning | Data Science

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SUMMARY

IT student and Python developer with a strong interest in Machine Learning and Data Science, and hands-on experience using NumPy, Pandas, and Scikit-learn for data analysis and model building. Completed a data science internship focused on data preprocessing, data visualization, and foundational machine learning concepts, and independently built end-to-end ML and full-stack projects covering recommendation systems, NLP-based text classification, and a multi-tenant Django platform with predictive analytics. Currently deepening expertise in Machine Learning, Artificial Intelligence, and data-driven technologies through continuous, hands-on learning. Open to internships and collaborative opportunities in AI, Machine Learning, and Data Science.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Machine Learning: Scikit-learn, PyTorch, Natural Language Processing (NLP)

Data Science: Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Training, Model Evaluation, Data Visualization

Libraries: Pandas, NumPy, Matplotlib, Seaborn, OpenCV

Web Technologies: Django, MERN Stack (MongoDB, Express.js, React, Node.js), HTML5, CSS3, Bootstrap, REST APIs

Web Scraping: BeautifulSoup, Requests, Selenium

Cloud & DevOps: AWS Lambda, Amazon EC2, Amazon Cognito, Git, GitHub

Tools & IDEs: Jupyter Notebook, VS Code, PyCharm

PROJECTS

RentMyLab - Multi-College Lab & Equipment Sharing Platform

- Architected and built an enterprise-grade, multi-tenant Django platform enabling students and staff across multiple colleges (LJ, SAL, SILV, GLS, LDC, NIRMA) to share and reserve laboratory space and equipment.
- Designed a role-based access control (RBAC) system with four distinct roles — Super Admin, College Admin, Lab Manager, and Student — each with tailored permissions, dashboards, and query scoping.
- Built a conflict-free slot booking engine with automated schedule validation, approval workflows, and a real-time check-in/check-out tracker with a 15-minute validation buffer.
- Developed explainable ML features using Scikit-learn: a rule-weighted lab recommendation engine, a Linear Regression model to predict equipment Remaining Useful Life (RUL), and a Logistic Regression model to flag high-risk booking cancellations.
- Implemented interactive analytics dashboards with Chart.js for peak usage hours, department booking distribution, and equipment status, plus JSON REST APIs for catalog and equipment data.

Tech Stack: Python, Django, Scikit-learn, Pandas, NumPy, SQLite, Bootstrap 5, Chart.js, REST APIs

Movie Recommendation System

- Built an end-to-end content-based movie recommendation system in Python, applying similarity analysis across genre, keyword, overview, cast, and director metadata to generate relevant recommendations.
- Performed data preprocessing, feature engineering, and text vectorization using CountVectorizer, then applied cosine similarity to rank and retrieve the top 5 most similar movies for a given title.
- Developed and deployed an interactive Streamlit web application that displays recommended movies with posters, ratings, and release year, retrieved dynamically through the TMDB API.
- Implemented secure API key handling, Git LFS for large file storage, and clear project documentation to support reproducible deployment.

Tech Stack: Python, Streamlit, Pandas, Scikit-learn, NLP, Cosine Similarity, TMDB API, Git LFS

SMS Spam Detection System

- Built a machine learning and NLP-based system to classify SMS messages as spam or ham (not spam), using a labeled SMS dataset.
- Applied text preprocessing techniques including tokenization, stopword removal, and stemming, followed by feature extraction using TF-IDF vectorization.
- Trained and evaluated a Multinomial Naive Bayes classification model, achieving 97.09% accuracy and 1.00 precision on the test set.
- Analyzed model performance using a confusion matrix and iterated on preprocessing and feature extraction to improve classification quality; built a real-time prediction interface for instant spam/ham classification.

Tech Stack: Python, Scikit-learn, NLP, TF-IDF, Multinomial Naive Bayes

EXPERIENCE

Data Science and Machine Learning Intern — 360 Design

Jun 2024 – Aug 2024

Ahmedabad

- Completed a 45-day internship focused on data science fundamentals and applied machine learning concepts.
- Gained hands-on experience in Python programming and exploratory data analysis.
- Worked on data preprocessing and cleaning tasks across small, real-world datasets.
- Learned and applied foundational machine learning concepts to sample datasets.

Website Design with PHP Intern — Patel Web Solution

Aug 2023

Ahmedabad

- Completed a 15-day internship in web development, building website designs using HTML, CSS, and basic PHP.
- Developed simple, functional web pages and forms for client-facing use cases.
- Learned the fundamentals of backend integration using PHP.

EDUCATION

Bachelor of Engineering, Information Technology

Jul 2025 – Sep 2028

L.J. Institute of Engineering and Technology

Diploma, Information Technology

Aug 2022 – Jun 2025

Government Polytechnic, Ahmedabad | CGPA: 8.61

CERTIFICATIONS

- Introduction to HTML, CSS, & JavaScript
- Cloud Knight Workshop – ACM PDEU
- Certificate of Participation in Idea Submission (Online) – Tic Tech Toe '26
- Exploratory Data Analysis for Machine Learning
- Certificate of Participation – tarkShaastra 2k26 Hackathon

HONORS & AWARDS

- Hackathon Winner – RE-INNOVATE: The Retail & Move the Market